Max Norris

Applied AI & Data Engineer

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⌖ Santa Cruz, CA

PROFILE

Engineer with 10+ years in aerospace (Northrop Grumman, Joby Aviation) moving into applied AI and data engineering. Has built and shipped LLM-powered products, including a live shopping-search agent with an evaluation harness and spending controls, and a daily satellite close-approach screening pipeline. Brings a decade of data-model, forecasting and standards-driven reporting work on safety-critical programs. Active DoD Secret clearance.

AI PROJECTS

ShopAI: LLM shopping-search agent *(live, 2026)*

- Shipped an agent that retrieves Google Shopping results, ranks them with Claude, and re-ranks on user ratings ("Shop Again"); Python service on Fly.io with Postgres, React/TypeScript front end on Vercel
- Built a live evaluation harness with an LLM judge (checked for self-agreement), regression cases, and cost and latency gates; 1,700+ automated tests
- Added per-session, per-visitor and daily spending caps so a public demo stays within budget

Satellite Point-of-View Tool *(live, 2026)*

- Daily screening of public tracking data for satellite close approaches using SGP4 propagation and scipy cKDTree spatial search; no LLM calls in the core
- Interactive 3D viewer (three.js) with Hipparcos and Gaia star catalogues

EXPERIENCE

March 2022 – Present
Senior / Lead Mass Properties Engineer, *Joby Aviation*

- Own weight and CG data for the S4 program: mass targets for each integrated product team, tracking, forecasting and reduction coordination
- Built the mass properties database and reporting process, implementing ANSI S-120A and SAWE RP 7/8
- Authored the Weight and Balance Manual and flight manual sections, and aircraft weighing procedures for flight test

January 2015 – March 2022
Mass Properties Engineer, *Northrop Grumman*

- Conceptual design focal point in Advanced Flight Sciences: parametric weight estimation models, MDO mass properties lead, contributions to multiple contract captures
- Detailed design weight and CG management, forecasting, reporting, and database development
- Production vehicle mass properties on F/A-18 and F-35A/B/C

SKILLS

- LLM Apps & Agents
- Evaluation & Testing
- Data Pipelines & Databases
- Technical Documentation

TOOLS

Python, TypeScript, React
Anthropic API, prompt design
PostgreSQL, pytest, GitHub Actions
Fly.io, Vercel
MATLAB, Phoenix Integration
CAD (NX / CATIA)

EDUCATION

Cal State Northridge
M.Sc. Physics, May 2014

Cal State Northridge
B.Sc. Astrophysics, May 2012

CLEARANCE

Active DoD Secret

LINKS

github.com/maxnorris01-bot